Adherent Mammalian Cell Subculture Protocol

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Safety statement: Prior to beginning an experiment, please read the Standard Operating Procedure (SOP) for this protocol as well as the Safety Data Sheets (SDS) for all chemical reagents listed. The SOP will contain a list of all chemicals for this protocol as well as directions for handling and disposal of hazardous wastes. It will also contain proper PPE guidelines and emergency protocols for accidents involving hazardous chemicals. Always be aware of your surroundings in the lab and be sure to communicate any hazards to those around you when performing an experiment.

Hazard statement: Any hazardous chemicals handled within this protocol will be marked with the following notation: (H). Please refer to the associated SOP to better understand the hazards and how to work with and dispose of these chemicals safely.

Background:
Mammalian cell culture has allowed for the study of various biochemical, biophysical, and genetic processes within the cell. By providing the right environment and nutrients, cells can multiply and thrive outside of an organism (i.e. ex vivo). The growth of cells in culture follows a standard pattern: a lag time after seeding (anywhere from 2-24 hours) is followed by a period of exponential growth called the "log phase." Cells will grow and divide until they either cover the entire surface area of the vessel or expend all the nutrients in the medium. At this point, their growth hits a plateau, so the cells need to be subcultured. Subculturing is the process of separating a small fraction of cells from a confluent population and propagating their proliferation. Theoretically, this allows for the culture to continue indefinitely. Cell lines can be either primary cultures or immortalized cultures. Primary cultures are cells taken directly from an organisms' tissues that have been treated with enzymes that break down excess connective tissue. They are not immortal and have a finite number of cell divisions before they senesce and apoptose. This allows scientists to harvest the cells and culture them, usually for direct studies of disease mechanisms. Immortalized cell lines can be derived from different sources: (1) primary cancer tissues that are isolated from people or animals, (2) cells that are not cancerous but have been genetically modified with active telomerase to keep telomeres long, or (3) by introducing viral oncogenes that are known to disrupt the cell cycle. Whichever way they are created, immortalized cell lines provide scientists with long-lasting specimens that can be expanded, frozen down, and perpetuated almost indefinitely. Cells in culture generally behave very consistently with regards to collecting data from assays, which makes them extremely attractive to use for experiments designed to uncover molecular mechanisms.

Equipment:

1. Cell culture vessels
2. Centrifuge
3. Serological pipettes – 5mL, 10mL, 25mL
4. Aspirating pipettes
5. Sterile centrifuge tubes
6. Incubator – set to 37°C and 5% CO_2
7. Cell counting device (we use the Countess)
8. Counting chamber slides

Reagents:

1. 1X Dulbecco's Phosphate Buffered Saline (DPBS)
2. 0.05% Trypsin-EDTA
3. Growth medium*
    a. Base medium (e.g. DMEM, RPMI, etc.)
    b. 10% fetal bovine serum (FBS)
    c. 1% antibiotic (usually penicillin-streptomycin combo)
4. Trypan Blue (H)

*Growth media varies widely in composition depending on the cell type and their specific requirements.

Procedural Overview:

In order to subculture cells, first the conditioned medium must be removed from the vessel. This is done with an aspirator connected to a waste container or with a pipette and waste container inside the Biosafety Cabinet (BSC). The cells are then washed with Dulbecco's Phosphate Buffered Saline (DPBS) to remove cell waste, metabolic debris, and the serum within the growth medium. It is essential that the DPBS lacks calcium and magnesium, as these ions inhibit trypsin, the enzyme that dissociates cells from the substrate. The serum in the medium also contains an enzyme called antitrypsin, which inactivates the trypsin enzyme. After a brief wash in DPBS, the cells are covered in trypsin, an enzyme that cleaves C-terminal peptide bonds on membrane proteins that keep the cells attached to the plate. The dissociation reaction is enhanced by incubation at 37°C. Trypsin is often combined with EDTA, an acid that chelates calcium and magnesium ions. As such, the cells lift from the substrate and become round. This is observed under a microscope. The dissociation reaction is quenched with complete growth medium since the serum contains antitrypsin. The cell suspension is then collected into a centrifuge tube and gently centrifuged to pellet the cells at the bottom of the tube. Centrifugation speeds and times vary depending on the cell line. Once pelleted, the supernatant is aspirated, and the cells are resuspended in DPBS. The volume used should allow for proper counting by the automated cell counter. If the cells are too dense, it can lead to inaccurate counting. Two counts are conducted and averaged before performing dilution calculations. After determining the volume of cells required for seeding, pre-warmed growth media is added to clean culture vessels before cells are added. The vessels are then shaken in a north-south, east-west motion to distribute the cells evenly across the culture vessel. Do not spin

the vessel in a circular motion; the centrifugal forces will concentrate the cells at the center. Cells are then placed in the incubator and grown until near confluence. Confluence is a qualitative description of the amount of surface area covered by cells. Typically, cells are grown to about 70-90% confluence before the next subculture.

Procedures:

1. Prior to subculturing, spray down the BSC with 70% ethanol diluted in deionized water (diH_2O).
2. Retrieve growth medium, 0.05% Trypsin-EDTA, and 1X DPBS from 4°C and place in a 37°C water bath for at least 15 minutes. Spray bottles down with 70% ethanol before placing into the BSC.
3. Retrieve cells from the incubator and observe under the microscope for proper morphology and any possible contamination.
4. Place cells in the BSC and set up the aspirator by taking a glass aspirating pipette and securing it to the rubber hose that connects to the waste receptacle. Turn the aspirator on by turning the vacuum nozzle counterclockwise.

BEFORE PROCEEDING: remember to spray gloved hands with 70% ethanol. Dry thoroughly before entering into the BSC.

5. Remove the conditioned medium from the cells by gently tilting the plate and aspirating all of the medium.
6. Pipette 1X DPBS over the cells and wash for approximately 10-15 seconds by gently tilting the PBS over the cell monolayer repeatedly. About 5-10 seconds is sufficient.
7. Aspirate the DPBS and pipette 0.05% Trypsin-EDTA over the cells. Use just enough to cover all the cells.
8. Incubate the Trypsin-EDTA on the cells at 37°C until a majority of cells have detached from the substrate. Use the microscope to confirm cells have detached. They will appear rounded and will slide when the vessel is tilted.
9. Pipette growth medium onto the cells to quench the Trypsin-EDTA. Collect cell suspension into a centrifuge tube using a pipette.
10. Centrifuge cells at 300xg* for 5 minutes. During this time, label the cell culture vessels that will be seeded and add the required amount of growth medium to each vessel.

*Centrifugation speeds and times will vary depending on the cell line.

11. Spray the centrifuge tube with 70% ethanol before placing it in the BSC. There should be a visible cell pellet at the bottom of the tube.
12. Aspirate the medium as close to the cell pellet as possible without disturbing the pellet.
13. Resuspend the cell pellet in an appropriate volume of 1X DPBS. The volume used should allow for accurate counting by the cell counter.
14. Remove an aliquot of cell suspension and place in a 0.6mL microcentrifuge tube.
15. Mix the aliquot of cells with an equal volume of Trypan Blue before pipetting into the counting slide. The Countess chamber slides hold 10µL per chamber.
16. Insert the slide into the machine and allow the machine to autofocus. Press 'Capture' and wait for the count to finish.

17. Record the live cell count and viability and discard the slide and tube into a hazardous waste container.
18. Calculate the volume of cells needed for each vessel according to the seeding density for the cell line being cultured (see table below for a reference). If you are unsure of how to seed a cell line, determine the seeding densities empirically by seeding multiple different densities in a multiwell plate and record the amount of time it takes for the cells to cover the surfaces of each well.

Cell Seeding Formula

$$\frac{seeding\ density\ (cells)}{cell\ count\ (\frac{cells}{mL})} = mL\ of\ cells\ that\ seed\ one\ culture\ vessel$$

19. Pipette the required volumes into the corresponding vessels and disperse the cells evenly within the vessels before placing back in the incubator. Use a north-south-east-west motion to disperse the cells. Never swirl the vessel in a circular motion, as this concentrates the cells at the center of the vessel.
20. Check the cells under the microscope to ensure they are dispersed, then return the vessel(s) to the incubator.

Cell Culture Vessel Reference Table

Vessel	Surface area (cm^2)	Seeding density*	Cell # at confluency*	Trypsin-EDTA (mL)	Working volume (mL)
60mm	21.5	$0.8x10^6$	$3.2x10^6$	1.5	5
100mm	56.7	$2.2x10^6$	$8.8x10^6$	3	10
150mm	145	$5x10^6$	$20x10^6$	7	25
6-well	9.6	$0.3x10^6$	$1.2x10^6$	1	2
12-well	3.5	$0.1x10^6$	$0.5x10^6$	0.5	1.5
24-well	1.9	$0.05x10^6$	$0.24x10^6$	0.2	0.5
48-well	1.1	$0.03x10^6$	$0.12x10^6$	0.1	0.3
96-well	0.32	$0.01x10^6$	$0.04x10^6$	0.05	0.1

*Numbers used are taken from HeLa cell studies. Every new cell line should be assayed for these parameters upon thawing, but after cryopreservation of low-passage cells.